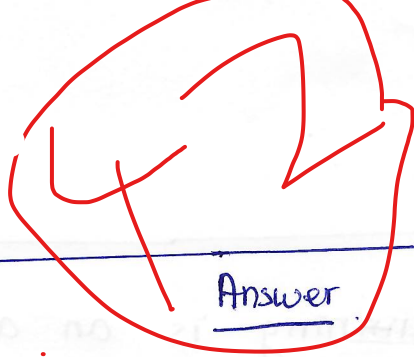




Answer

1. D ✓

2. C ✓

3. A ✓

4. C ✓

5. B ✓

6. A ✓

7. ~~A~~ A ✓

8. C ✓

9. D ✗

10. B ✓

11. B ✓

12. C ✓

13. B ✓

14. C ✓

15. D ✗

16. C ✓

17. B ✓

18. C ✓

19. B ✓

20. C ✗

21. Supervised machine learning is an algorithm that was trained on label data and try to predict the label of new input.

22. ML model seem to have good accuracy with training data but it has low accuracy with testing data, we can say that this model is overfitting, we need to do something to make it more generalize.

23. Regression analysis is a technique that we use to find the relationship between feature.

24. two Challenges of machine learning algorithm

- Accuracy: ~~the model need to perform well~~
- Resource: Computation Resource needed to make a better model
- Data: we need to collect proper data in order to make our model better.

25. two major application of Data Science

- House price prediction (Regression)
- Bad news classification (classification)

26. Difference between Structure and Semi-Structured Data.

- structured Data: it is properly store in a specific format like table
- Semi-Structured: Data was stored in json or xml.

27. To Find outlier in Dataset there are multiple techniques
- plot data in Scatter plot
 - plot data in Box plot
 - See the Distribution of the Data.
28. Data pre-processing is a technique or step in order to process and transform the data before fitting it to a machine learning algorithm.
29. In KNN data needs to be normalized because KNN calculates the distance between data points and classifies the data point into a specific class based on its neighbours. If we don't normalize data, the model will be biased towards a specific feature.
30. We can discretize the continuous attribute by setting the range of each class. Example: Age: 5-10 (Kids), (10-18) adult
31. K is the number of centroids or clusters in K-means.
32. Few hyperparameters in machine learning: gradient descent, adam,
33. Gradient Descent is an algorithm that uses machine learning models in order to minimize the cost by adjusting the weights.

33. Gradient descent is an hyperparameter tuning ~~technique~~ technique that ~~used~~ used in machine learning that try to optimized the cost by adjusting the weight.

34. Comment on SVM classifier

→ The First model seem to have a good performane because it is properly Classify the data and the distane between boudary line is large enough

→ The Second model performe not well because it ~~missed~~ ~~classified the class~~ the data not properly was overlap with the boundary line.

→ The Third model seem to be overfitting because the distance between the boundary line is small.

35. Radial Basic function is SVM. work by calculate the distance between boundary to point and it's respetive class.